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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
IV Semester B. E. Regular Examinations SEP/OCT – 2024
Electronics and Communication Engineering
SIGNALS AND SYSTEMS

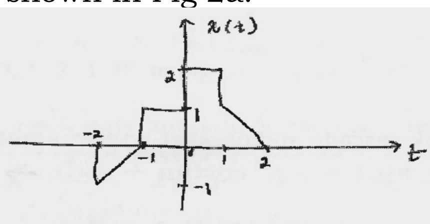
Time: 03 Hours ***Maximum Marks: 100***

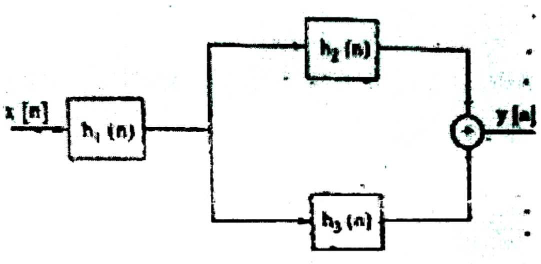
Instructions to candidates:
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

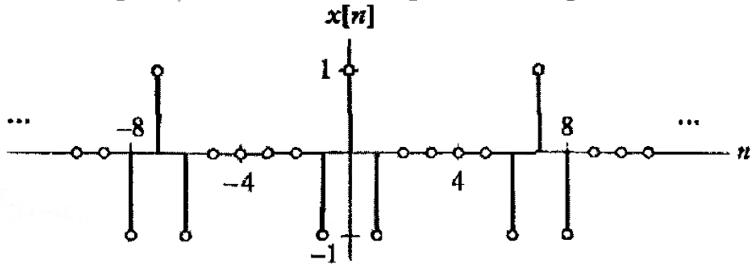
PART-A

			M	BT	CO
1	1.1	The energy of the following signal $x(t) = e^{-at}u(t)$ for a>0 is ______.	1	2	1
	1.2	If the impulse response of the system is given as $h(t) = e^{-\|t\|}$. the step response of the system is ______.	2	1	2
	1.3	If $x(n) = \{1, 2, 3, 4, 5, 6, 0, 0\}$, then circularly shifted signal $2x(n-1) =$ ______.	1	2	2
	1.4	If the $DTFT$ of $x(n)$ is $X(e^{j\Omega})$ and $DTFT$ of $x'(n)$ is $X(e^{j(\Omega-a)})$, then $x'(n) =$ ______.	2	3	2
	1.5	Consider the following dataset $x = [1, 1, 1, 1, 1, 1, 1, 1]$. The value of skewness is ______, and it indicates ______ distribution.	2	2	1
	1.6	The difference in the number of complex multipliers required for 32-point DFT and 32-point radix-2 FFT is ______.	1	3	3
	1.7	The first 4 points of a 6-point DFT of a real valued sequence are $2,\ 1-j3, 2+j5, 3+j4$. Let $X4, X5$ be fifth and sixth points in DFT then $\|X5\|/\|X4\| =$ ______.	2	3	3
	1.8	Consider two 16-point sequence x[n] and h[n]. Let the linear convolution on x[n] and h[n] be denoted by y[n], while z[n] denotes the 16-point inverse discrete Fourier transform ($IDFT$) of the product of the 16-point $DFTs$ of $x[n]$ and $h[n]$. The value(s) of k for which $z[k] = y[k]$ is ______.	2	3	1
	1.9	Let $x(n) = (1/2)^n u(n), y(n) = x^2(n)$, and $Y(e^{j\Omega})$ is the Fourier transform of $y(n)$, then $Y(e^{j0}) =$ ______.	2	2	4
	1.10	If $y(t) = x(t) + x(kt) - x(9t) + x(t-1)$ then, for what value of k the system be time invariant.	1	1	4
	1.11	Let $x(n) = \{1,2,0,3,-2,4,7,3\}$. then $\sum_{k=0}^{7}\|X[k]\|^2 =$ ______.	2	3	3
	1.12	A 5-point sequence x(n) is given as $x[-3] = 1, x[-2] = 1, x[-1] = 0, x[0] = 5$ and $x[1] = 1$. Let $X(e^{j\Omega})$ denoted the discrete-time Fourier transform of $x(n)$. The value of $\int_{-\pi}^{\pi} X(e^{j\Omega})d\Omega$ is ______.	2	3	4

PART-B

2	a	Consider the signal shown in Fig 2a.  Fig.1.a Plot the signals (i) $x(2-t)$ (ii) $x(2t-1)$ (iii) $x(4-\frac{t}{2})$	06	3	1

	b	Sketch $x(t) = -4u(t+4) - 3u(t+1) + 5u(t-1) + 2u(t-3)$	06	3	1
	c	Find out the energy and power of (i) $x(t) = u(t)$ and (ii) $x(t) = r(t)$	04	2	2
3	a	For the following impulse response $h(t)$ given, determine whether the corresponding system is (i) Memory less, (ii) Casual and (iii) Stable? $h(t) = e^{-2t}u(t-1)$ $h(t) = \cos(\pi t)u(t)$	06	2	1
	b	Evaluate $y[n] = x[n] * h[n]$, where $x[n]$ and $h[n]$ are shown in Fig 3.b  Fig. 3.b.	06	3	2
	c	Evaluate the step response of the LTI system represented by the impulse response: $h[n] = \delta(n) - \delta(n-2)$	04	4	2
		OR			
4	a	Find the convolution of $x_1(n) = \left(\frac{1}{3}\right)^n u(n)$ with $x_1(n) = (2)^n u(-n)$	05	3	3
	b	Consider the interconnection of LTI systems shown in Fig.4b. Find the overall impulse response if $h_1(n) = (1/4)^n u(n+1)$, $h_2(n) = 2\delta(n)-3\delta(n-1)$ and $h_3(n) = u(n-1)$.  Fig 4.b	06	2	3
	c	The output of a discrete-time system is related to its input $x[n]$ as follows: $y[n] = 5\,x[n] + 2\,x[n-1] + 3\,x[n-2] + x[n-4]$ i) Formulate the operator for the system relating y[n] to x[n]. ii) Develop the block diagram representation for H, in both cascade and parallel implementation.	05	4	2
5	a	Let the input to a discrete time LTI system be $x(n) = 8 + \cos(\pi n+\frac{\pi}{6})$. Determine the output of this system for the impulse response, $h(n) = \frac{\sin\left(\frac{\pi}{8}n\right)}{\pi n}$. Also, sketch $X(e^{j\Omega})$, $H(e^{j\Omega})$ and $Y(e^{j\Omega})$.	10	4	4
	b	Let p(t) be aperiodic continuous time signal defined by $p(t) = 5cos(900\pi t) + 6cos(500\pi t)$, is sampled with a sampling frequency $f_s = 1\,kHz$. Calculate the $DTFS$ coefficient of the sampled signal $p(n)$.	06	3	3
		OR			

6	a	Find the $DTFS$ representation of the periodic signal shown in Fig 6.a  Fig 6.a	08	3	3
	b	Find the DTFT representation for the following periodic signal $$x[n] = \cos\left(\frac{\pi n}{8}\right) + \sin\left(\frac{\pi n}{5}\right)$$	08	2	3
7	a	Let x(n) be a finite length sequence with its 4-point DFT, $X(k) = \{0, 1+3j, 1, 1-3j\}$. Using the properties of DFT, find the $DFTs$ of the following sequences: i) $d_1(n) = e^{j0.1\pi n}.x(n)$ ii) $d_2(n) = \cos(0.25\pi n).x(n)$ iii) $d_3(n) = x((n-1))_4$	06	3	3
	b	Compute 4-point circular convolution of $x(n) = \{1, 5, -2, 3\}$ and $h(n) = \{1, -1, 2, -2\}$ using FFT algorithm.	10	2	2
		OR			
8	a	Compute circular cross correlation of following sequence using DFT and IDFT: $x(n) = \{1, 4, -2, 7\}$ and $h(n) = \{1, 5, -2\}$	06	2	3
	b	Find the output y(n) of a filter whose impulse response $h(n) = \{2, 2, 2\}$ and input signal, $x(n) = \{3, -1, 0, 1, 3, 2, 0, 1, 2, 1, 0\}$ using overlap-add method. Apply 5-point circular convolution.	10	4	4
9	a	How can time domain and frequency domain features be combined to improve the accuracy of signal classification illustrate with an example.	08	3	4
	b	What the significance of the mean and variance in the context of time domain signal analysis. How does it help in characterizing the signal?	08	2	3
		OR			
10	a	What role does feature selection play in the classification of signals? How can irrelevant or redundant features impact the performance of a classification system?	08	2	4
	b	Consider the dataset representing signal values over time, and the values are $[22, 25, 28, 18, 34, 30, 200, 18, 15, 150, 122, 231, 11, 67]$. Find the outliers in the given data set using IQR method.	08	4	4